DC-DC CONVERTOR BOARD
TECHNICAL SPECIFICATION

TABLE OF CONTENTS

1 GENERAL

SRS function is to drive the elevator to the nearest landing during power failure, SRS is designed such that it can replace EBD, ERD as whole unit. It can be interfaced with MX/NMX machine (with LWD).

This document describes the technical specifications of the DC-DC convertor board(5KW) which is part of SRS system.

1.1 SYSTEM ARCHITECTURE

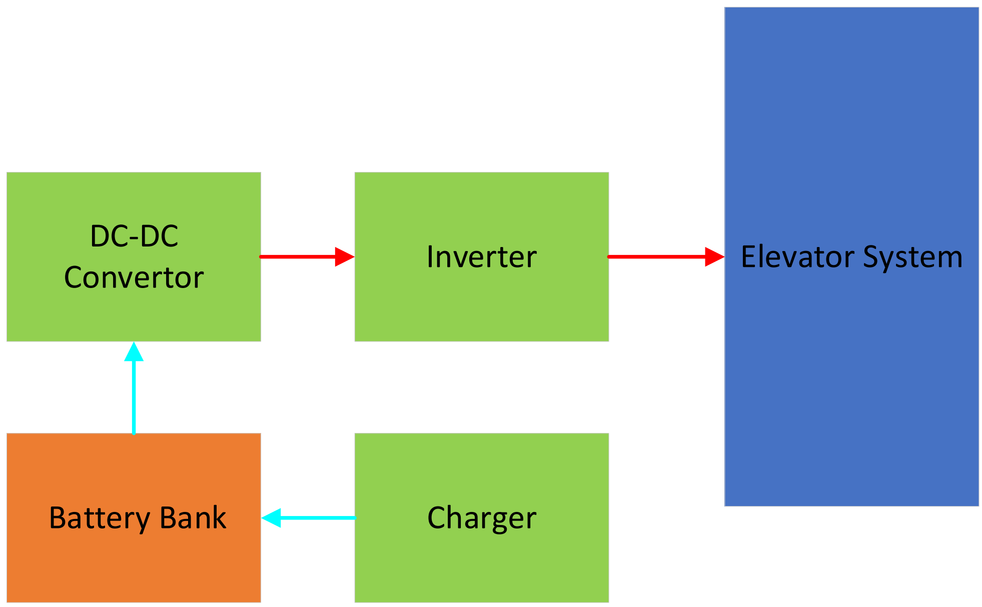


Figure 1:System Architecture

1.2 COMPONENT TECHNOLOGY

The component technology decided to use is specified in chapter Manufacturing process.

2 FUNCTIONAL OPERATION

2.1 Requirement Specification

Parameter	Value
Output power rating	5KW
Output voltage	380 VDC
Input voltage max	42 V DC
Input voltage min	36 VDC
Controller type	Digital
Communication interface	RS485,RS232
Board Size	155MMX215MM

2.2 DC-DC Converter Protection

- Temperature compensation for charger.
- Output overload protection
- Battery overload and fire protection
- Battery charger short circuit protection
- Battery reverse polarity protection
- Thermal protection for Transformer & Power device’s
- Instantaneous over current protection for MOSFET’s

2.3 BLOCK DIAGRAM

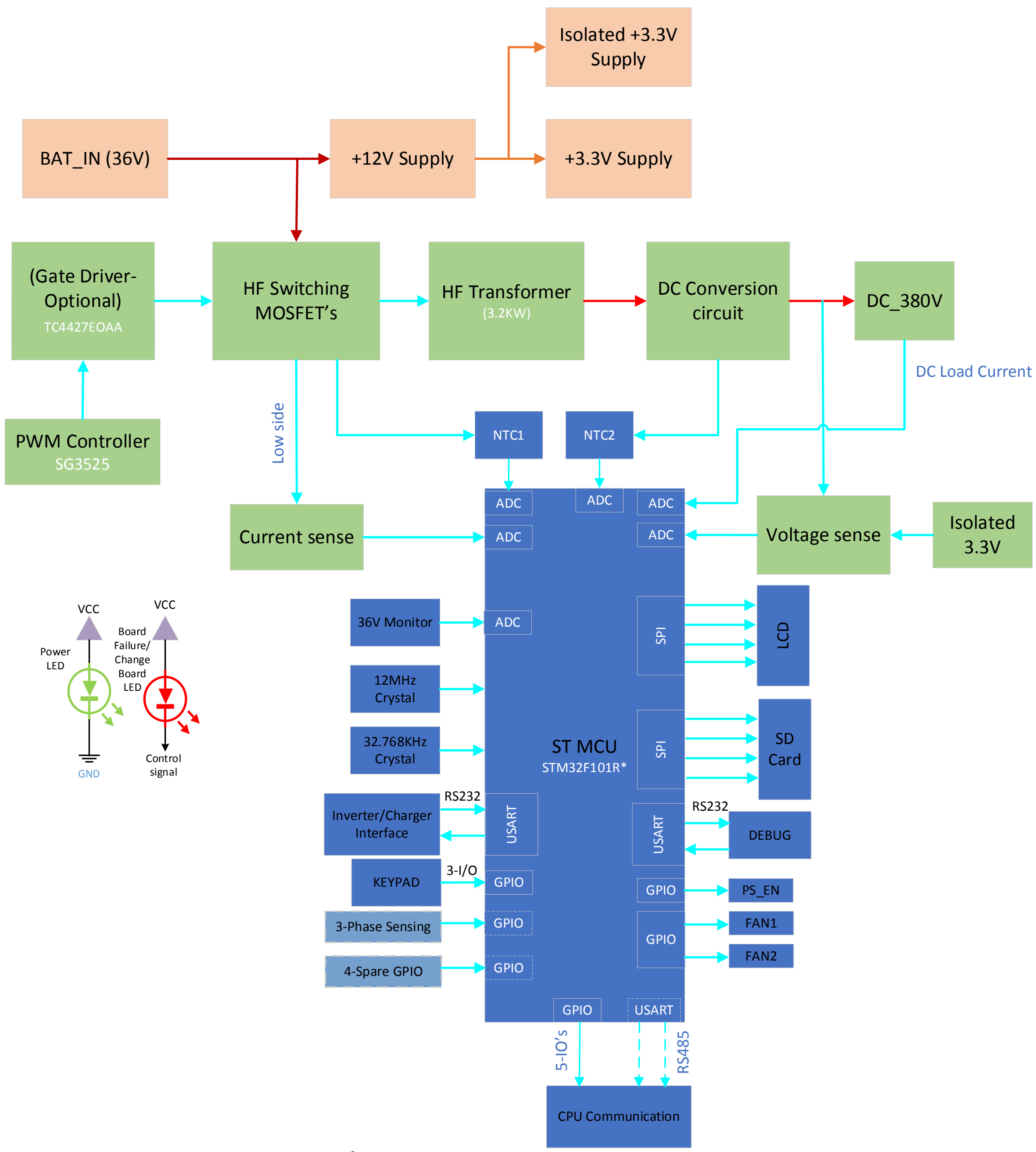


Figure 2:DC-DC Converter Board Block Diagram

DRAFT

2.4 DESCRIPTION OF SIGNALS AND SYMBOLS

Signal or Symbol	Description
PWM	PULSE WIDTH MODULATION
HF	HIGH FREQUENCY

2.5 PWM SWITCHING CONTROLLER

The SG3525 is the PWM controller which is used in the Board for h-Bridge MOSFET switching purpose. This IC's supports oscillator frequency range of 50Hz to 400KHz.The required operating range for this Design is 43KHz.

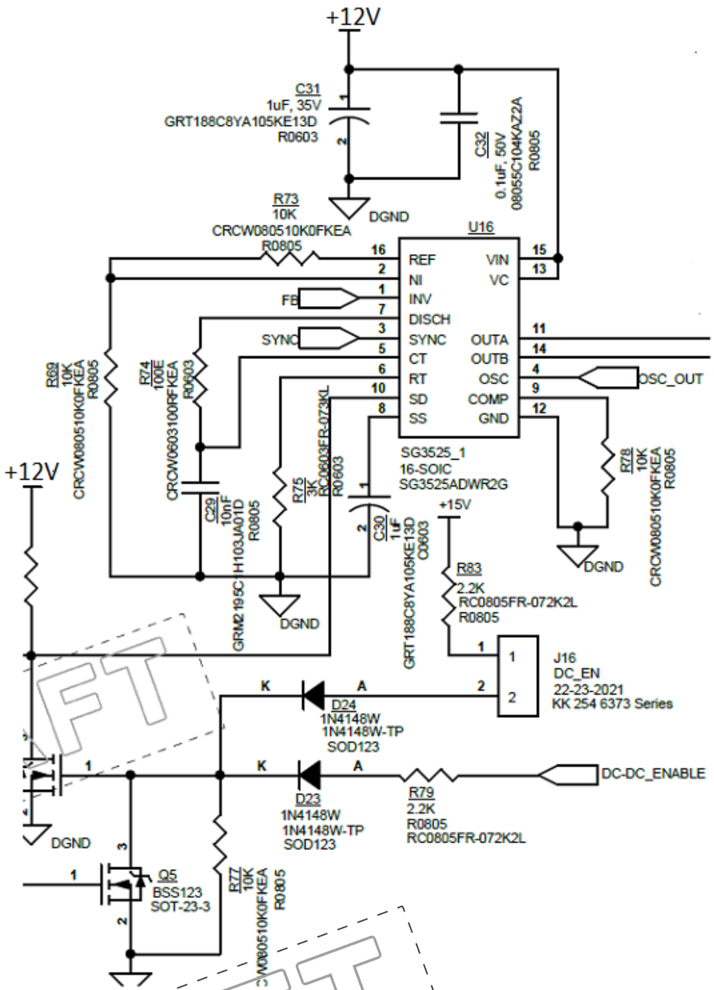


Figure 3: PWM controller Circuit

2.6 POWER MOSFET DRIVER

Output of PWM controller is given to Dual High speed MOSFET Driver IC(TC4427EOAA) to drive the push pull topology MOSFET's.

TC4427EOAA Gate driver is capable of peak output current 1.5A with Matched Rise, Fall and Delay Times.

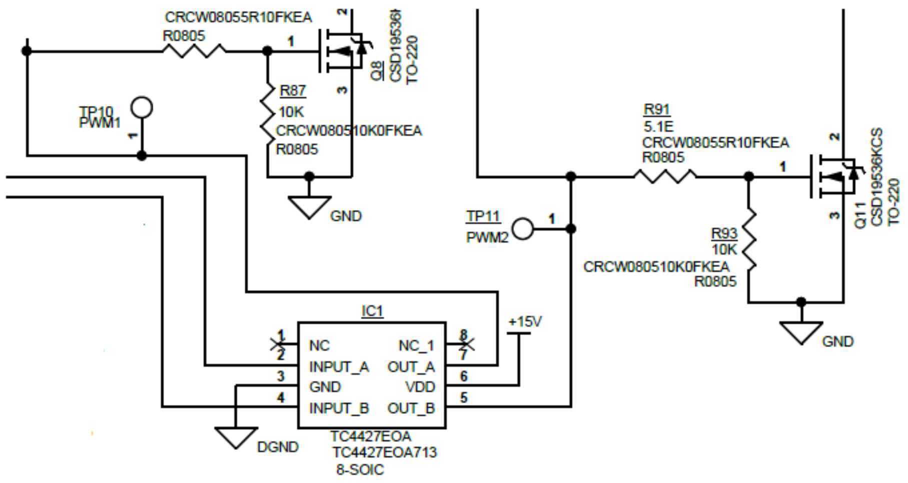


Figure 4: MOSFET DRIVER

Output of Driver IC is connected to the gate terminals of the MOSFET's(CSD19536KCS). Whereas the MOSFET drain terminal are connected with the High frequency transformer secondary side.

2.7 HF TRANSFORMER

The 3.2KW transformer is used for stepping up the output to 380V level. Below are the specification details of the HF transformer.

36V transformer Specification.xlsx

2.8 CURRENT SENSE

The current sense is done at the lower arm of MOSFET in terms of voltage drop across the shunt resistor connected at the Lower side of MOSFET. And that output is given to MUC through OPAMP(voltage follower).

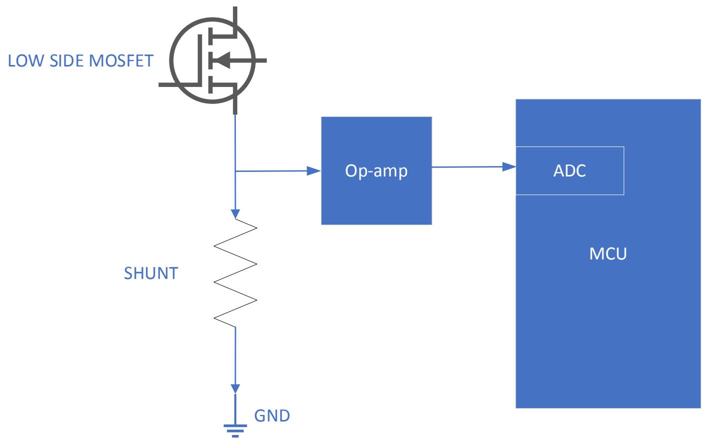


Figure 5: Current Sense

2.9 DC VOLTAGE SENSE

The rectifier output voltage is sensed by MCU through isolated opamp(AMC1302) followed by a voltage divider resistors.

A separate +3.3V supply is used for the opamp (mean-well).

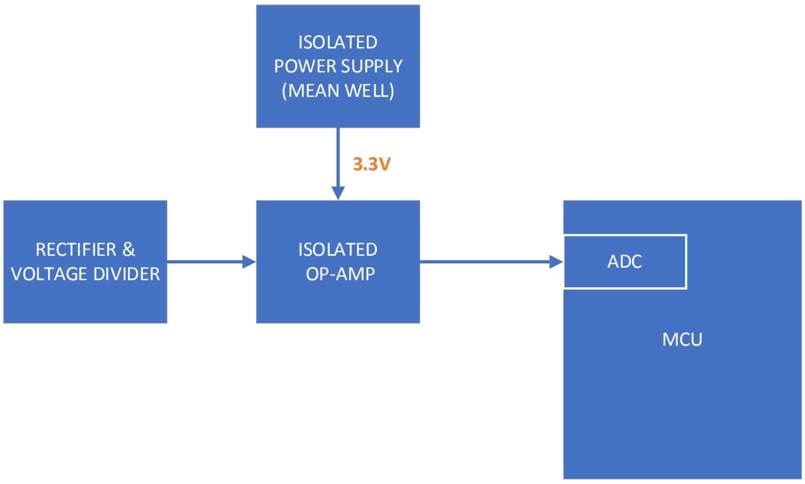


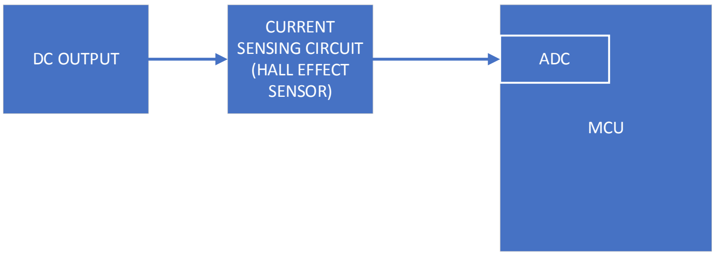
Figure 6: DC Voltage Sense

2.10 DC LOAD CUREENT SENSE

The load current is sensed and feedback to the MCU. Which is done based on the Hall effect sensor on the DC output(Load) side.

This output is given to ADC channel of MCU.

DC LOAD CURRENT SENSING

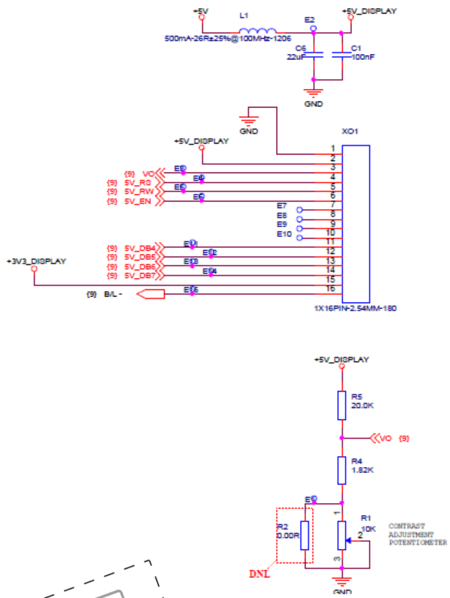


2.11 LCD DISPLAY-PARALLEL

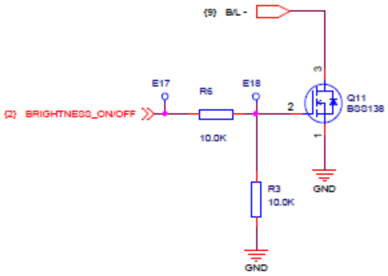
This LCD used for setting parameters and to displays fault codes. And it operates at 3.3VThe LCD module uses the 8 bit for data communication from Micro controller. However this LCD module can also work with 4 bit data (upper bit:DB4-DB7) communication, hence in this PCBA 4 bit is used for data (upper bit:DB4-DB7) communication.

(5V Supply to be changed as 3.3V)

CONNECTOR FOR GLCD / LCD

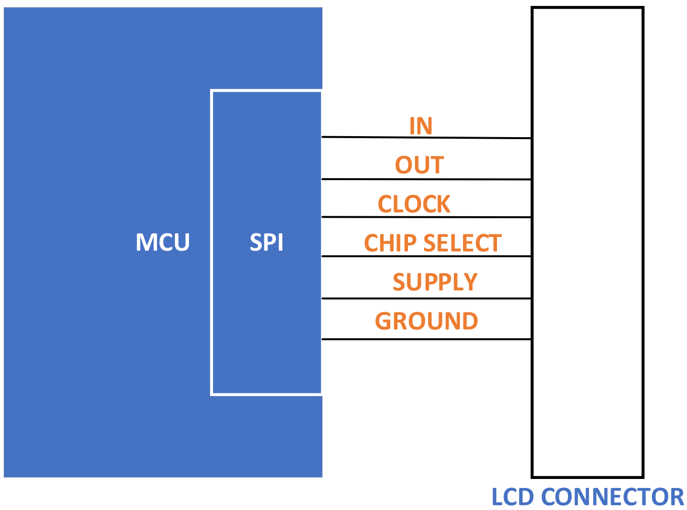


This LCD will be switch ON/OFF by using the MOSFET (Q11). Below the control signal circuit.



2.11.1 LCD SERIAL

The LCD interface is designed with serial communication for Future purpose. By using SPI channel the LCD interface can be done. For this purpose, SPI channel retained. SPI supported LCD with 3.3V supply to be used in this.



2.12 INVERTER/CHARGER PCBA INTERFACE

From DC-DC Converter PCBA, the RS-232 interface is used for communicating with Inverter and charger PCBA's.

Below things are communicated for Inverter board

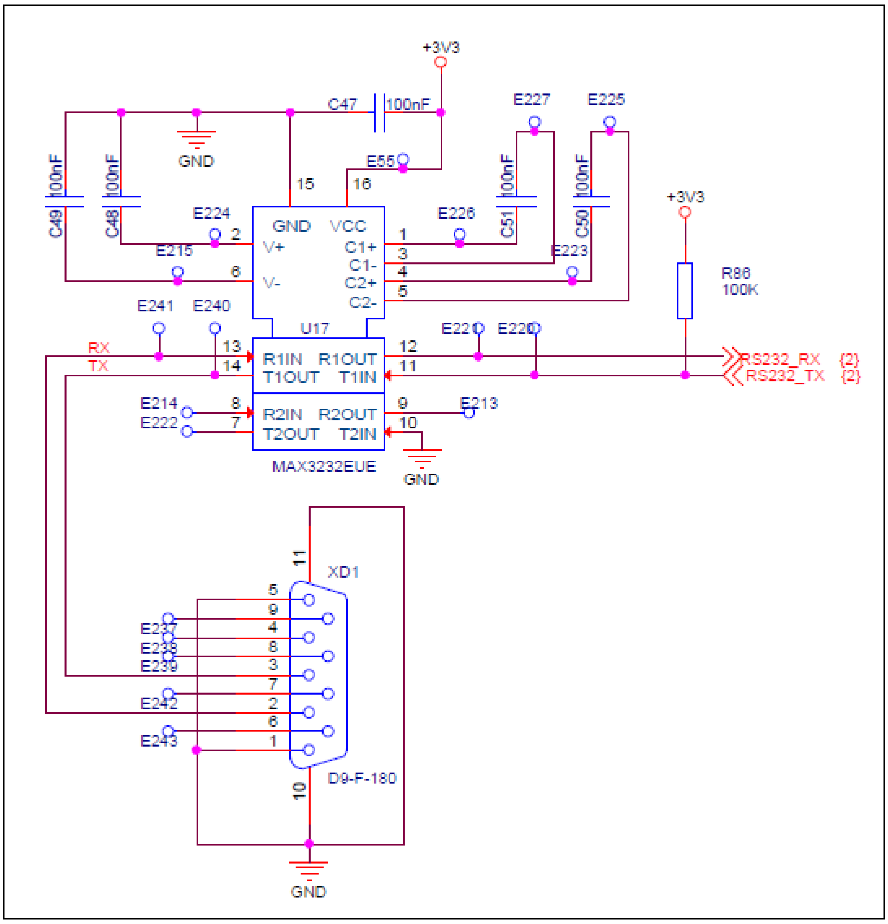
1. **Enable /Disable:** This signal is output signal which is Enable/disable the Inverter PCBA.
2. **Inverter Failed/Change Board:** This signal is input signal for this PCBA which is used to indicate the Inverter PCBA failure and inform to change the PCBA.
3. **Inverter Mode:** This signal is input signal for this PCBA which is used to indicate the Inverter PCBA status.

Below things are communicated for charger board

1. From Inverter PCBA, the RS-485 interface for communicate with charger PCBA for Battery charging status, Battery condition and fault codes etc.

MAX3232 RS-232 Transceiver IC is used for this interface. This transceiver supports and capable of full-duplex mode and data transfer rate of 120Kbps.

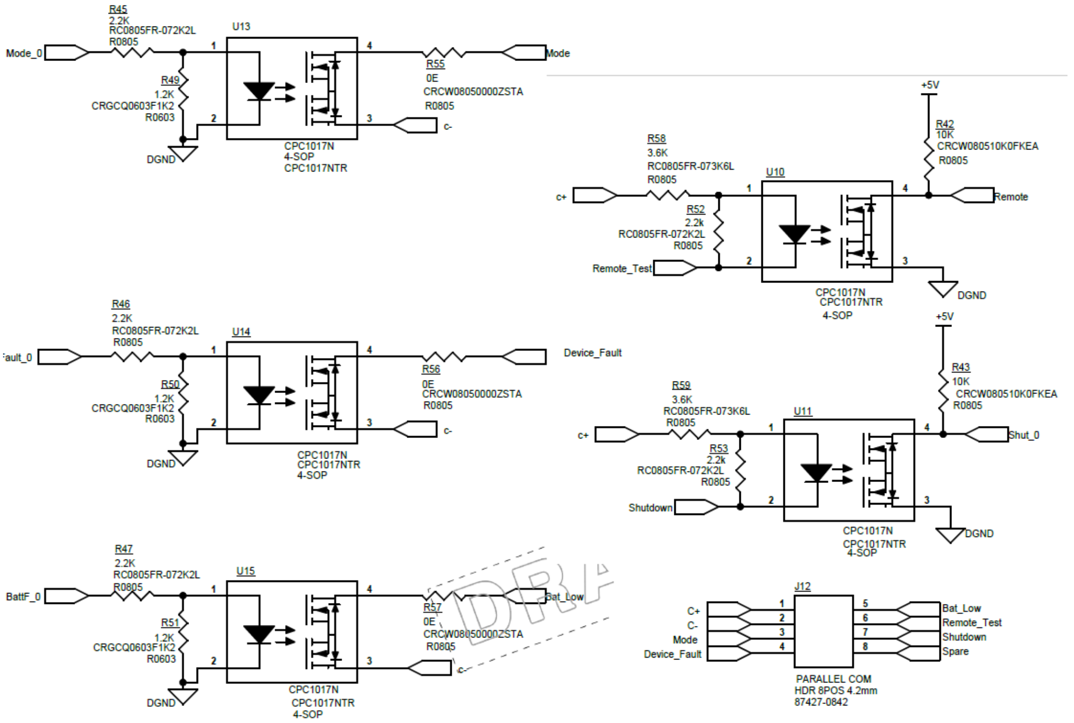
Refer the circuit below for RS232 Interface Section.



2.13 ELEVATOR CPU INTERFACE-PARALLEL

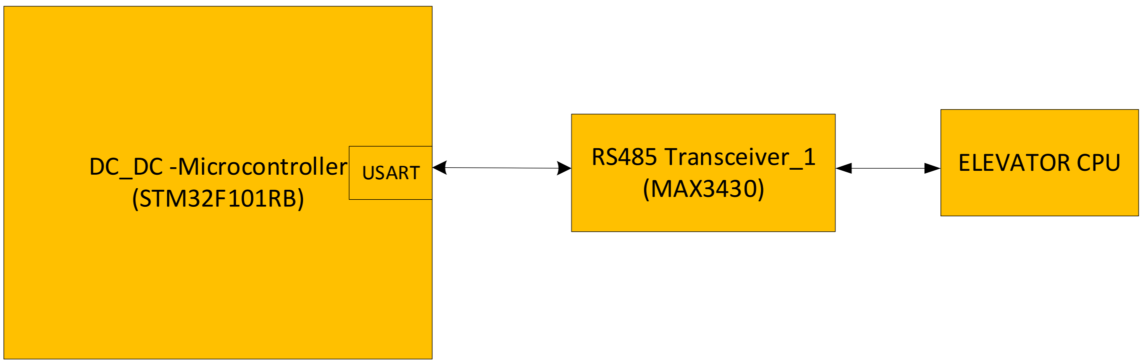
DC-DC board uses the below signals for CPU Controller communication,

1. Mode: Mode of running which means, Main mode or rescue mode will communicate to CPU
2. Device_Fault: This signal will communicate to CPU that this Device is fault.
3. Bat_Low: This signal indicates when the Battery voltage is lower than 38V(YTD) the unit will not go to ERD mode.
4. Remote_Test: this signal will communicate Remote test.
5. Shutdown: This signal will go LOW and will make the unit come out of ERD mode



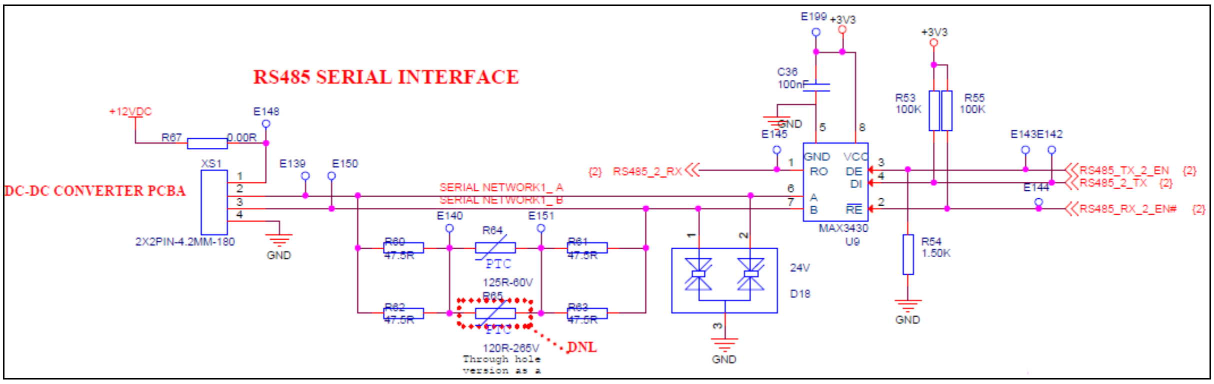
2.14 ELEVATOR CPU INTERFACE-SERIAL

This RS485 Interface used for DC-DC board to Communicate with CPU through serial for future purpose.



The MAX3430 fault-protected RS-485 transceiver features protection from over voltage signal faults on communication bus lines. Messages between DC TO DC CONVERTER, Elevator CPU shall be communicated to through RS 485 interface. Baud rate for this communication will be 115200kbps. It has,

- ±80V Fault Protection
- ±12kV ESD Protection
- 250kbps Data Rate
- Allows Up to 128 Transceivers on the Bus
- -7V to +12V Common-Mode Input Voltage Range

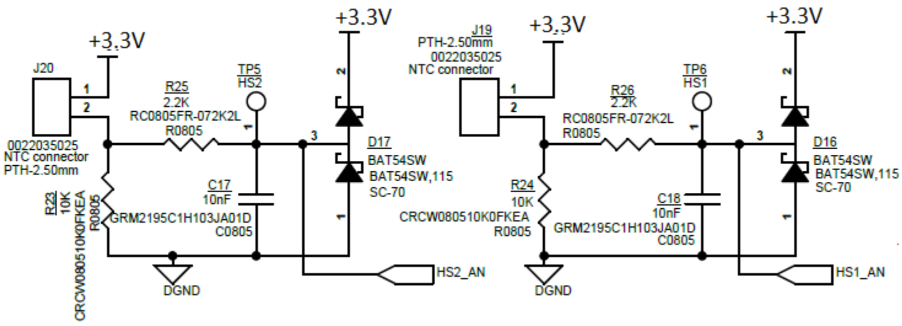


2.15 NTC’s and FAN’s

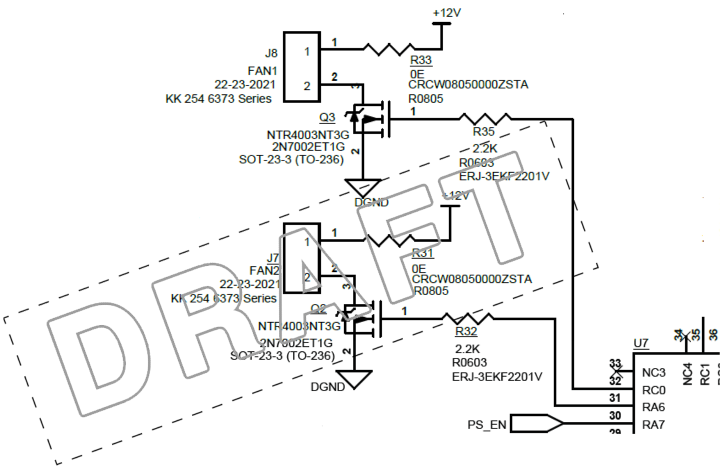
Two NTC’s used in this mode for monitoring Temperatures of MOSFET’s and rectifier diodes which are given to ADC channel of MCU.

These temperature sensors are mounted on the heatsinks of MOSFET and Diode. These outputs will be used by MCU to trun ON the FAN’s when heats up.

Below the NTC Interface circuit diagram

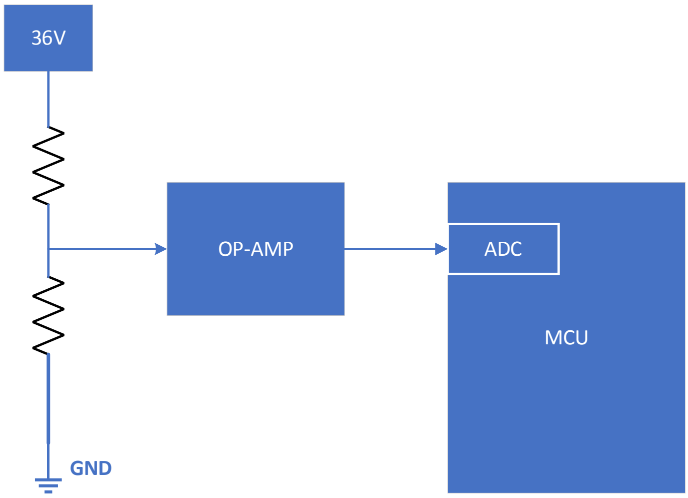


Below the FAN Interface circuit diagram



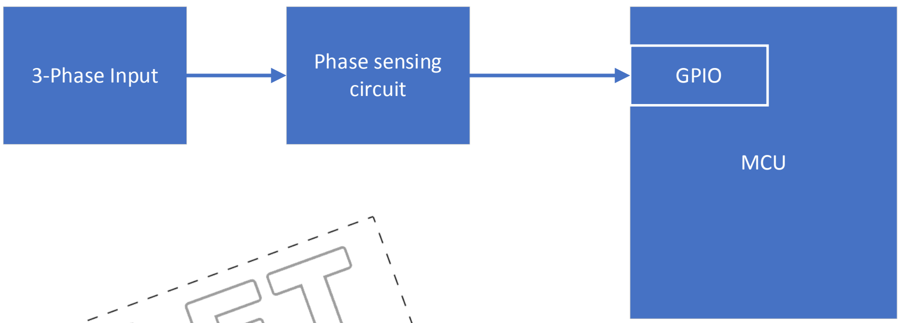
2.16 +36V MONITORING

The board Input voltage 36V is monitored in MCU. This is done opamp followed by voltage divider. Here opamp is used as voltage follower and which is used for better output accuracy.
The output of opamp is given to the ADC channel of MCU.

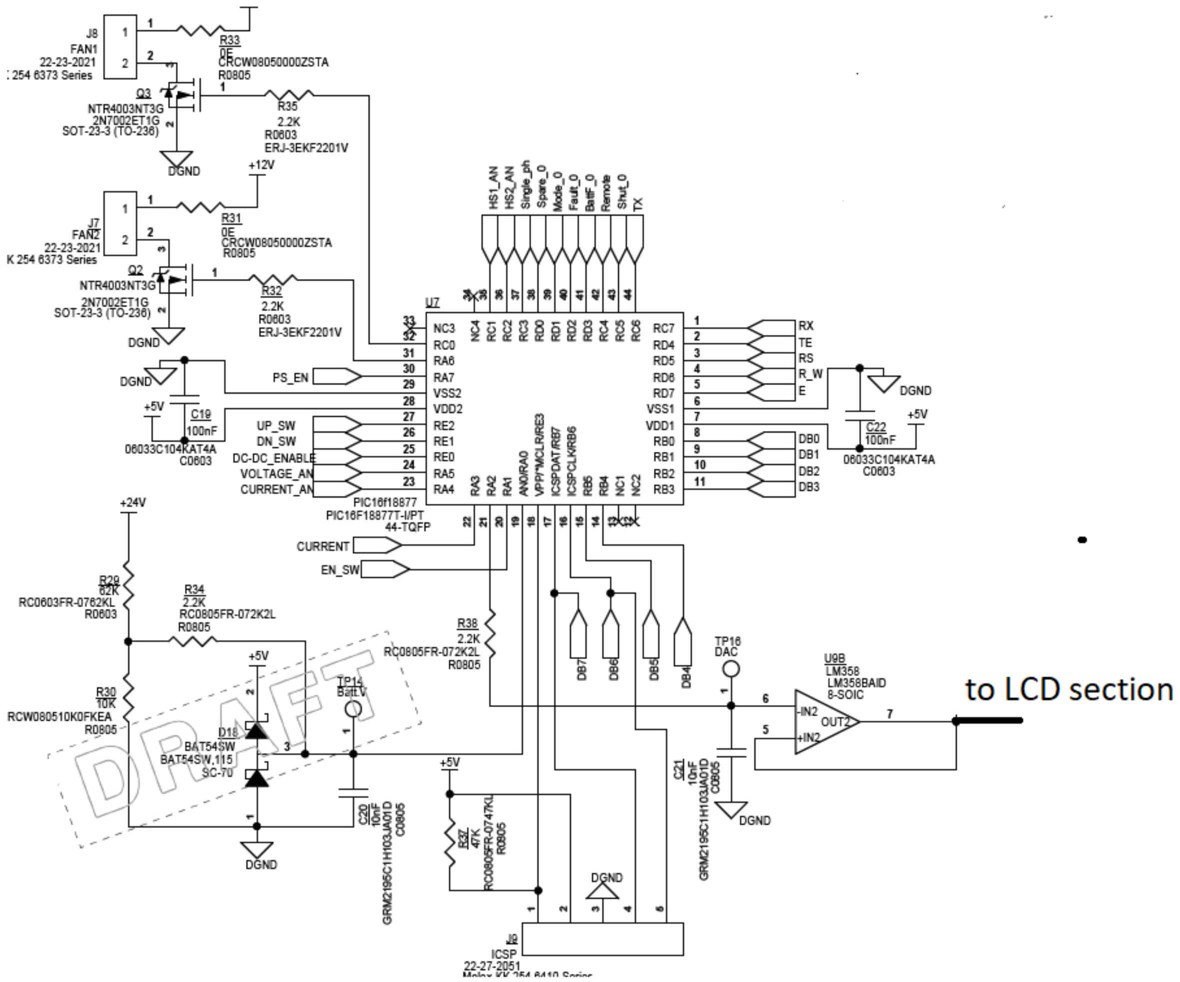


2.17 3-PHASE SENSING

3-phase sensing to be added for future purpose. This can be not assembled in current design.
Output of the 3-phase sensing circuit is given to one of ADC channel in MCU.



2.18 MICROCONTROLLER SECTION



(Above image has PIC MCU these signals to be maintained and MCU to be changed as STM).

The STM32F101RB(medium-density access line family incorporates the High-performance ARM Cortex™-M3 32-bit RISC core operating at a 36 MHz frequency, High-speed embedded memories (Flash memory up to 128 Kbytes and SRAM up to 16 Kbytes), and an extensive range of enhanced peripherals and I/Os connected to two APB buses. All devices offer standard communication interfaces (two I2Cs, two SPIs, and up to three USARTs), one 12-bit ADC and three general-purpose 16-bit timers. The STM32F101xx medium-density access line family operates in the –40 to +85 °C

2.18.1 TIMERS

There are three synchronically general-purpose timers embedded in the STM32F101xx Medium-density access line devices. These timers are based on a 16-bit auto-reload Up/down counter, a 16-bit presale and feature 4 independent channels each for input capture, output compare, PWM or one pulse mode output. This gives up to 12 input captures / output compares / PWMs on the largest packages. The general-purpose timers can work together via the Timer Link feature for synchronization or event chaining. Their counter can be frozen in debug mode. Any of the general-purpose timers can be used to generate PWM outputs. They all have independent DMA request generation. These timers are capable of handling quadrature (incremental) encoder signals and the digital outputs from 1 to 3 hall-effect sensors.

2.18.2 INTERNAL MEMORY

EMBEDDED FLASH AND SRAM

The ARM Cortex™-M3 processor is the latest generation of ARM processors for embedded systems. It has been developed to provide a low-cost platform that meets the needs of MCU implementation, with reduced pin count and low-power consumption, while delivering outstanding computational performance and an advanced system response to interrupts. The ARM Cortex™-M3 32-bit RISC processor features exceptional code-efficiency, delivering the high-performance expected from an ARM core in the memory size usually associated with 8- and 16-bit devices. The STM32F103xx performance line family having an embedded ARM core is therefore compatible with all ARM tools and software.

EMBEDDED FLASH MEMORY

128 Kbytes of embedded Flash is available for storing programs and data.

EMBEDDED SRAM

16 Kbytes of embedded SRAM accessed (read/write) at CPU clock speed with 0 wait states.

2.18.3 INTERNAL WATCHDOG

INDEPENDENT WATCHDOG

The independent watchdog is based on a 12-bit down counter and 8-bit prescaler. It is clocked from an independent 40 kHz internal RC and as it operates independently of the main clock, it can operate in Stop and Standby modes. It can be used either as a watchdog to reset the device when a problem occurs, or as a free-running timer for application timeout management. It is hardware- or software-configurable through the option bytes. The counter can be frozen in debug mode.

WINDOW WATCHDOG

The window watchdog is based on a 7-bit down counter that can be set as free-running. It can be used as a watchdog to reset the device when a problem occurs. It is clocked from the main clock. It has an early warning interrupt capability and the counter can be frozen in debug mode.

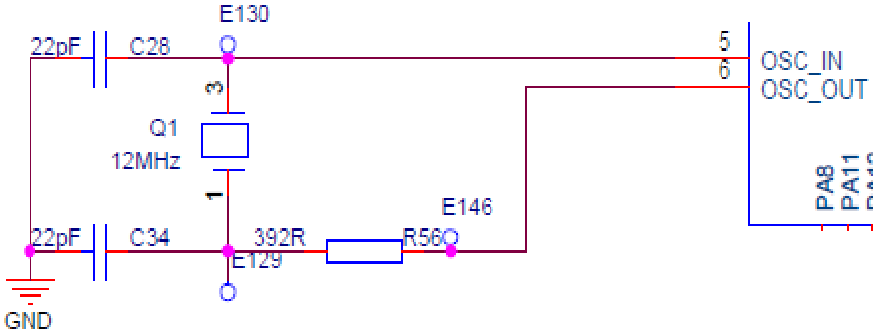
2.18.4 IO MAPPING(TBD)

Pin no	Pin function	Type	Assigned net	Default /programmed state
1	VBAT	Supply	3.3V	
2	PC13	Output		
3	PC14	Output		
4	PC15	Input		
5	PD0	Input	OSC _IN	
6	PD1	Output	OSC_OUT	
7	NRST	Input	RESET BY JTAG	
8	PC0	Output		
9	PC1	Output		
10	PC2	Input		
11	PC3	Input		
12	VSSA	Supply	ANALOG GND	
13	VDDA	Supply	3.3V ANALOG	

14	PA0/WKUP	Input		
15	PA1	Input		
16	PA2	Output		
17	PA3	Input		
18	VSS_4	Supply	GND	
19	VDD_4	Supply	3.3V	
20	PA4	Output		
21	PA5	Output		
22	PA6	Output		
23	PA7	Input		
24	PC4	Output		
25	PC5	Output		
26	PB0	Output		
27	PB1	Output		
28	PB2	Output		
29	PB10	Input		
30	PB11	Input		
31	VSS_1	Supply	GND	
32	VDD_1	Supply	3.3V	
33	PB12	Output		
34	PB13	Output		
35	PB14	Output		
36	PB15	Output		
37	PC6	Output		
38	PC7	Output		
39	PC8	Output		
40	PC9	Output		
41	PA8	Output		
42	PA9	Output		
43	PA10	Input		
44	PA11	Input		
45	PA12	Input		
46	PA13	Bidirectional	JTMS	
47	VSS_2	Supply	GND	
48	VDD_2	Supply	3.3V	
49	PA14	Output	JTCK	
50	PA15	Input		
51	PC10	Output		
52	PC11	Input		
53	PC12	Output		
54	PD2	Input		
55	PB3	Output		
56	PB4	Output		
57	PB5	Output		
58	PB6	Output		
59	PB7	Output		
60	BOOT0	Input		
61	PB8	Output		
62	PB9	Input		
63	VSS_3	Supply	GND	

64	VDD_3	Supply	3.3V	

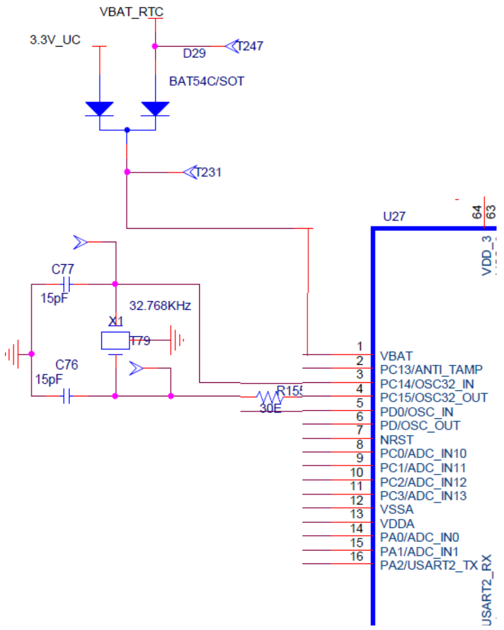
2.19 MAIN CRYSTAL



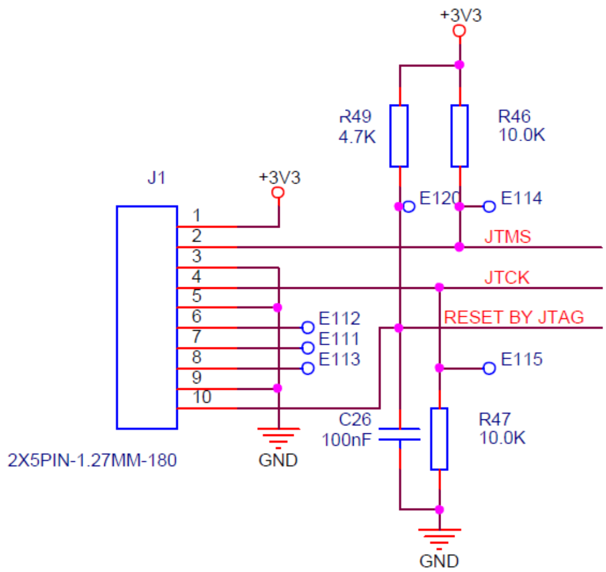
12 MHz crystal is used to generate clock signals.

2.20 CRYSTAL FOR RTC

32.768KHz is used for the real time clock purpose.



2.21 JTAG INTERFACE

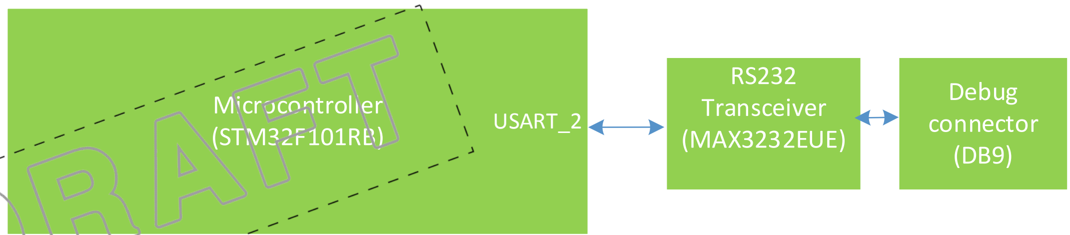


The ARM SWJ-DP Interface is embedded and is a combined JTAG and serial wire debug port that enables either a serial wire debug or a JTAG probe to be connected to the target. The JTAG TMS and TCK pins are shared with SWDIO and SWCLK, respectively, and a specific sequence on the TMS pin is used to switch between JTAG-DP and SW-DP.

2.22 DEBUG -RS232 SERIAL INTERFACE

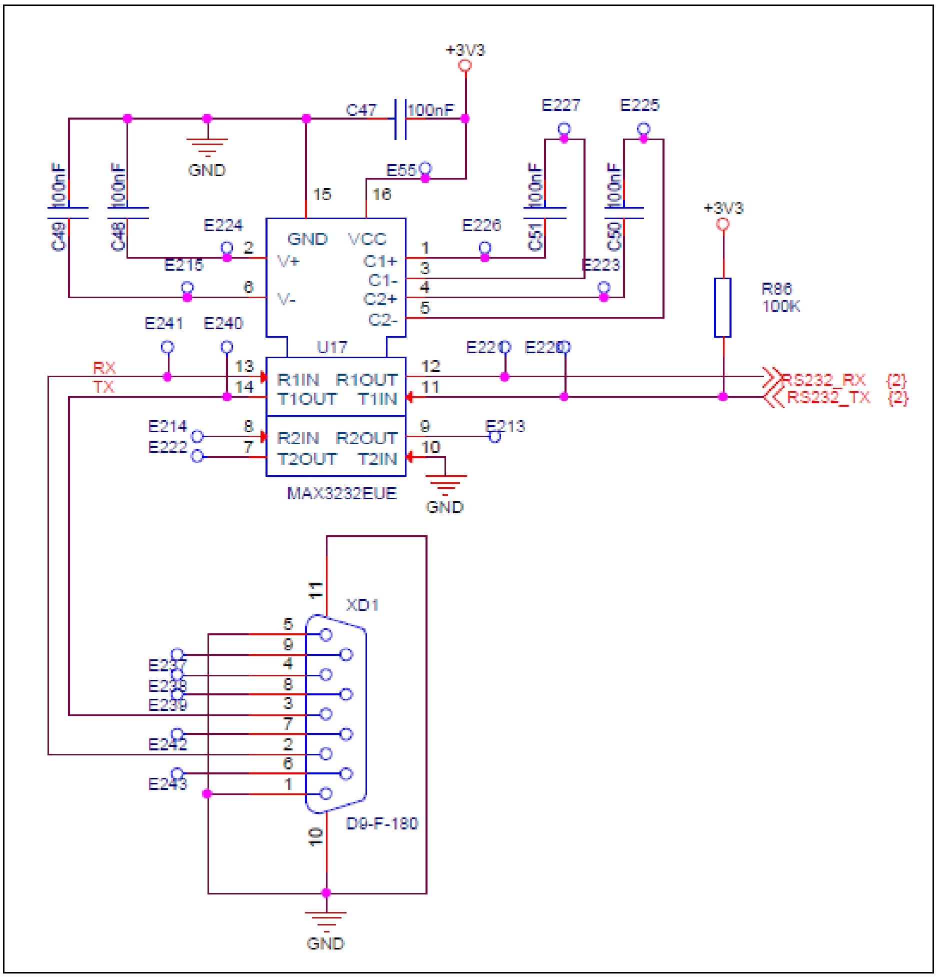
The Debug Unit (DBGU) provides a single entry point from the processor for access to all the debug capabilities of the processor. Port details are shown in below table

Signal	I/O Line	Peripheral
RS232_RX	PA3	USART2
RS232_TX	PA2	USART2

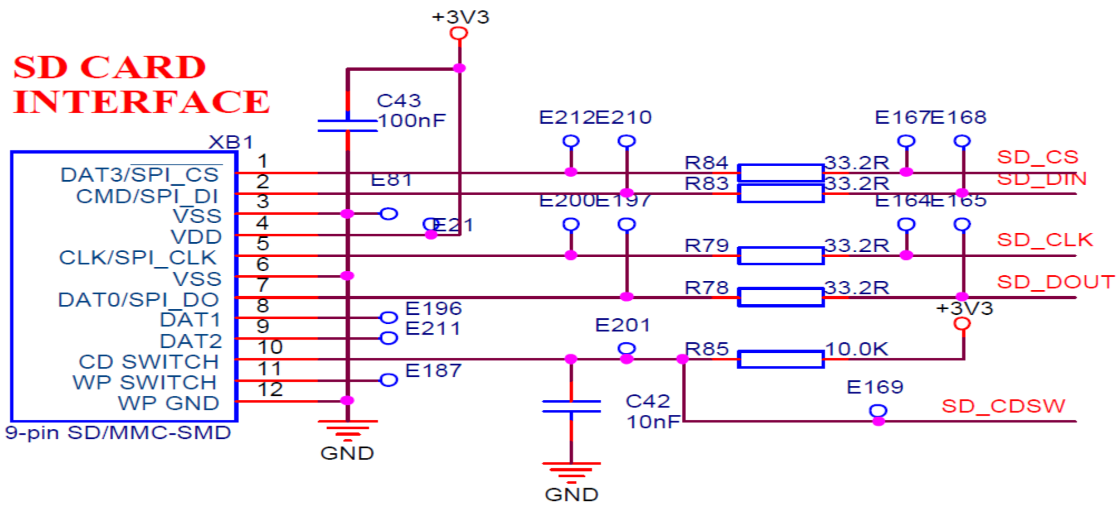


MAX3232 RS-232 Transceiver IC is used for this debug interface. this transceiver is supports and capable of full-duplex mode and data transfer rate of 120Kbps.

Refer the circuit below for RS232 Debug Interface Section.


+3V3
C47
100nF
E227
E225
GND
E55
15
16
C49
100nF
C48
100nF
C51
100nF
C50
100nF
E224
E226
GND
VCC
V+
C1+
C1-
C2+
C2-
V-
E215
E223
+3V3
R86
100K
E241
E240
U17
E221
E220
RX
TX
R1IN
R1OUT
T1OUT
T1IN
RS232_RX
RS232_TX
{2}
E214
E222
R2IN
R2OUT
T2OUT
T2IN
E213
MAX3232EUE
GND
XD1
E237
E238
E239
E242
E243
D9-F-180
GND

2.23 SD CARD INTERFACE

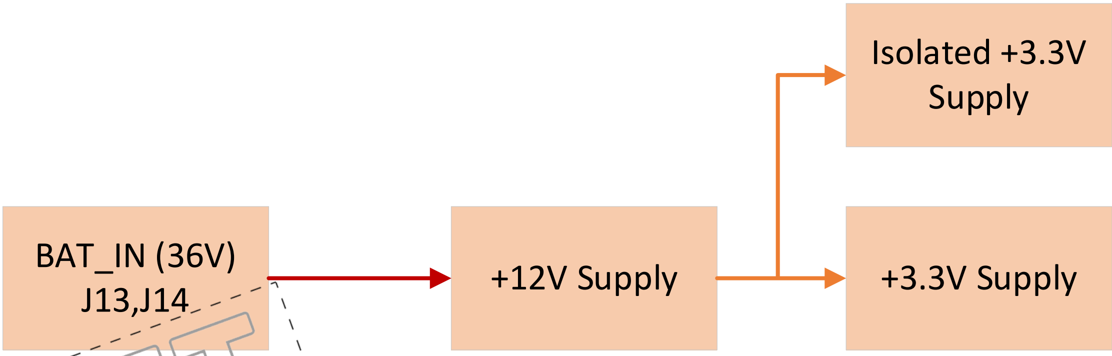


SD card interface used to load the application software in the field.

2.24 POWER SUPPLY

Input for DC-DC converter Board is +36VDC from battery, which is given @ J13,J14 Connectors.
This 36V will be converted to +12V by a Switching regulator (LM5085_DGK_8).
Then +12V will be converted into +3.3V using another switching regulator (LM2596S-3.3).

A separate 3.3V power supply module(mean well or similar TBD) is used for the voltage sense opamp.

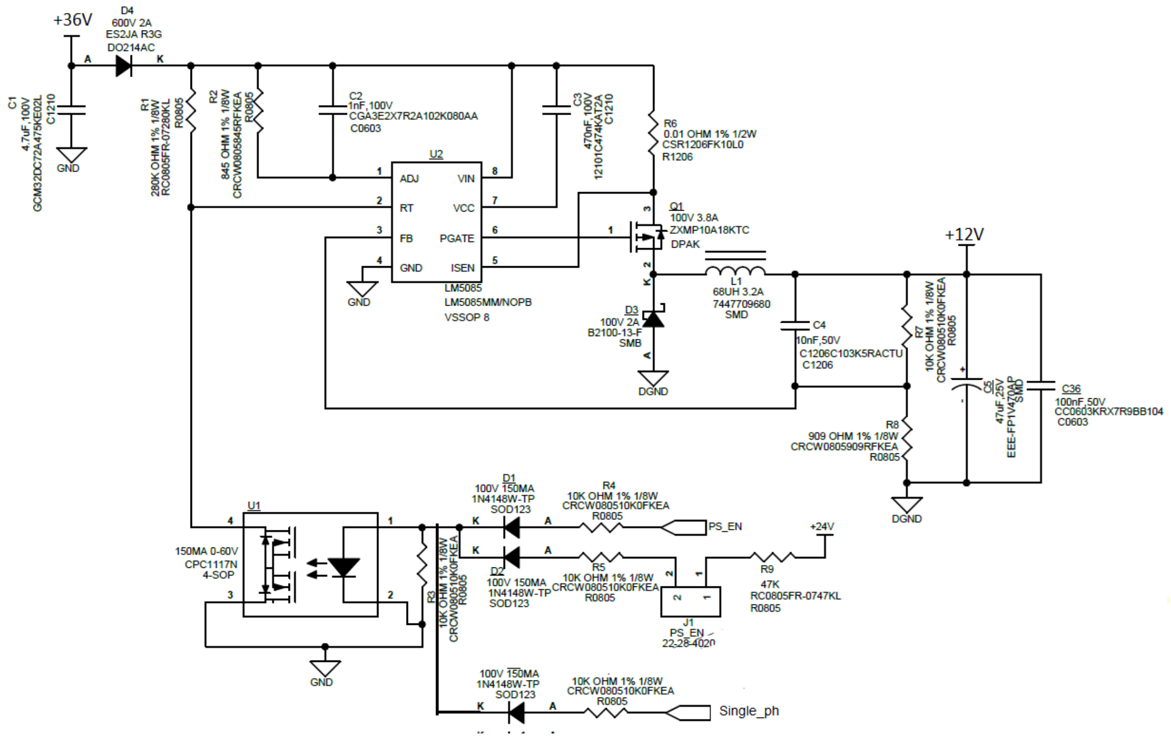


The power section will have to meet the below requirements.

- Input voltage min/max 36V/42V
- Input current min/max ?
- Output current min/max ?
- Protection→ Over Temperature, Short circuit, over current.

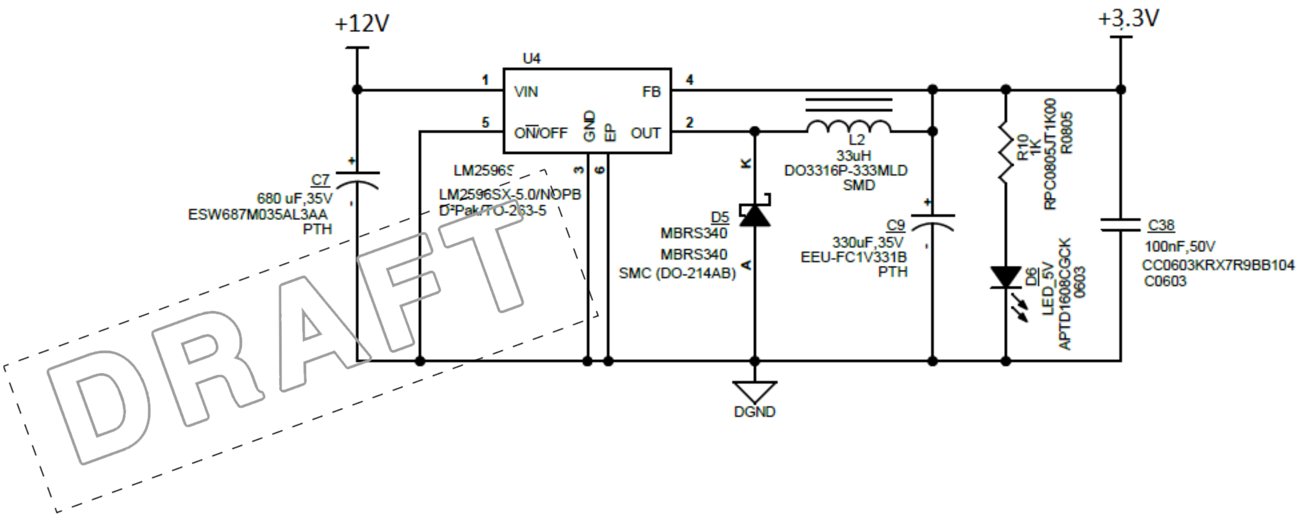
2.24.1 +12V POWER SUPPLY

Below +12V section reference circuit, Values are mentioned for 24V input and 15V out, that all to be changed for 36V input and 12V output



2.24.2 +3.3V POWER SUPPLY

Below +3.3V section reference circuit, this 3.3V is used as supply for MCU,LCD,MAX IC's and OP-AMPs



2.24.3 +3.3V POWER SUPPLY MODULE

SPAN02A-03 Power supply model is used to supply the isolated error amplifiers in the board.

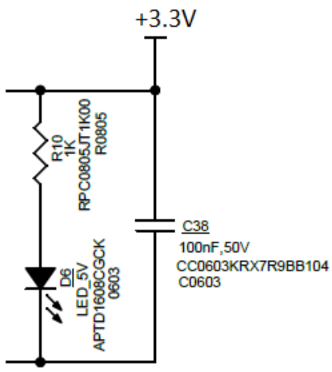
SPAN02-03 is 2W isolated and regulated module type DC-DC converter with SIP8 package. It features high temperature range -40 to +90 high isolation voltage and short circuit protection. Which support 12V input and can deliver 3.3V output with low leakage current.

Below table shows the pin out of the power supply model

Pin-Out		
Pin No.	SPAN02 (Single output)	DPAN02 (Dual output)
1	-Vin	-Vin
2	+Vin	+Vin
3	R.C.	R.C.
5	N.C.	N.C.
6	+Vout	+Vout
7	-Vout	Common
8	N.C.	-Vout

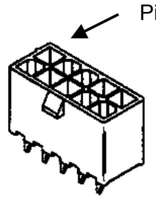
2.24.4 POWER OK INDICATORS

The green LEDs are connected to the +3.3V, power outputs to indicate the state of the power supply .

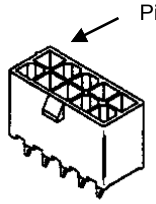


3 INTERFACES

3.1 ELECTRICAL INTERFACE

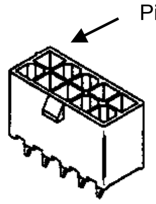
Position number	J13	Pin 1  Note: Pin number is not same
Connector type	9 Pin Shank Terminal Through Hole	
Name	DC Input	
From/To	Battery /DC-DC board	

Pin	Signal name	Type	Voltage Min	Voltage Max	Current Min	Current Max	Low max High min	Notes
1	36VDC		36V DC	42 V DC				

Position number	J14	Pin 1  Note: Pin number is not same
Connector type	9 Pin Shank Terminal Through Hole	
Name	DC Input	
From/To	Battery /DC-DC	

Pin	Signal name	Type	Voltage Min	Voltage Max	Current Min	Current Max	Low max High min	Notes
1	Bat-0V DC			0V DC				

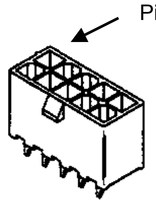
Position number	J15
Connector type	9 Pin Shank Terminal Through Hole
Name	Transformer Input connector
From/To	Centre tap of Transformer



Note: Pin number is not same

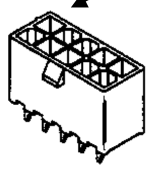
Pin	**Signal name**	**Type**	**Voltage Min**	**Voltage Max**	**Current Min**	**Current Max**	**Low max High min**	**Notes**
1	0VDC			24 V DC				

Position number	J17
Connector type	9 Pin Shank Terminal Through Hole
Name	Transformer Input connector
From/To	Centre tap of Transformer

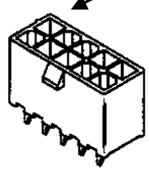


Note: Pin number is not same

Pin	**Signal name**	**Type**	**Voltage Min**	**Voltage Max**	**Current Min**	**Current Max**	**Low max High min**	**Notes**
1	24VAC			24 V AC				

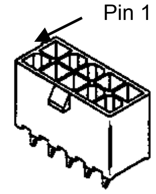
Position number	J18	Pin 1  Note: Pin number is not same
Connector type	9 Pin Shank Terminal Through Hole	
Name	Transformer Input connector	
From/To	Centre tap of Transformer	

Pin	Signal name	Type	Voltage Min	Voltage Max	Current Min	Current Max	Low max High min	Notes
1	24VAC			24 V AC				

Position number	J4	Pin 1  Note: Pin number is not same
Connector type	TERM BLOCK HDR 3POS VERT 5.08MM	
Name	DC Output	
From/To	DC-DC Board /Load	

Pin	Signal name	Type	Voltage Min	Voltage Max	Current Min	Current Max	Low max High min	Notes
1	380VDC			380VDC				
2	GND							

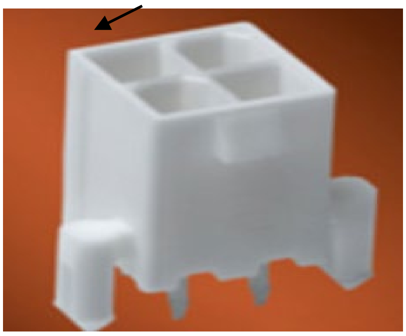
Position number	J16
Connector type	CONN HEADER VERT 2POS 2.54MM
Name	DC_EN
From/To	PWM CONTROLLER ENABLE/DISABLE



Note: Pin number is not same

Pin	Signal name	Type	Voltage Min	Voltage Max	Current Min	Current Max	Low max High min	Notes
1	12V			12VDC				
2	GND							

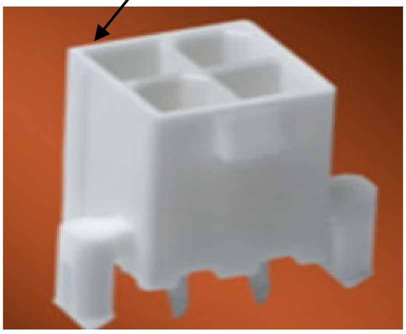
Position number	XS1
Connector type	MINI FIT CONNECTOR 2X2PIN-4.2MM-180 (KM261998)
Name	RS485 CONNECTOR
From/To	DC-DC CONVERTER TO INVERTER/CHARGER



Note: Pin number is not same

Pin	Signal name	Type	Voltage Min	Voltage Max	Current Min	Current Max	Low max High min	Notes
1	+12VDC	Supply		12VDC	500mA			
2	Serial network A	In/Out	0VDC	3.3VDC	6mA	10mA		
3	Serial network B	In/Out	0VDC	3.3VDC	6mA	10mA		
4	GND	Supply	0VDC		500mA			

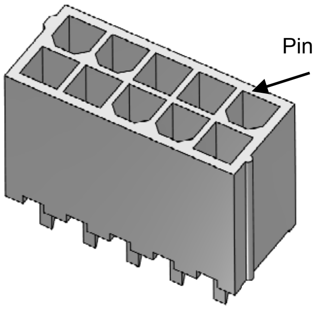
Position number	XS2
Connector type	MINI FIT CONNECTOR 2X2PIN-4.2MM-180 (KM261998)
Name	RS485 CONNECTOR
From/To	RS485 SPARE CONNECTOR



Note: Pin number is not same

Pin	**Signal name**	**Type**	**Voltage Min**	**Voltage Max**	**Current Min**	**Current Max**	**Low max High min**	**Notes**
1	+12VDC	Supply		12VDC	500mA			
2	Serial network A	In/Out	0VDC	3.3VDC	6mA	10mA		
3	Serial network B	In/Out	0VDC	3.3VDC	6mA	10mA		
4	GND	Supply	0VDC		500mA			

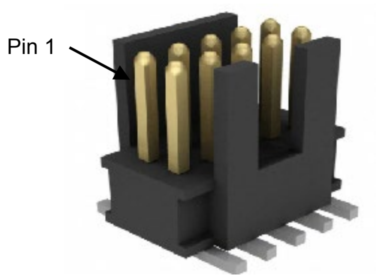
Position number	J16
Connector type	MINIFIT CONNECTOR 2X4PIN-4.2MM-180
Name	CPU CONTROLLER SIGNALS
From/To	From CPU to DC-DC



Note: Pin number is not same

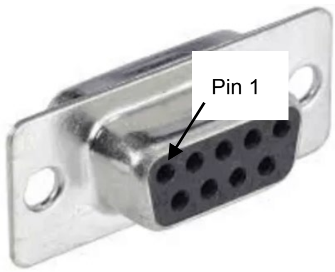
Pin	**Signal name**	**Type**	**Voltage Min**	**Voltage Max**	**Current Min**	**Current Max**	**Low max High min**	**Notes**
1	Common + (12V)	Power		12VDC		10 mA		
2	Common -	Gnd		12VDC		10 mA		
3	Mode	Input		12VDC		10 mA		
4	Device_Fault	Input		24VDC		10 mA		
5	Battery Low	Gnd		24VDC		10 mA		
6	Remote Test	Input		24VDC		10 mA		
7	Shutdown	Input		24VDC		10 mA		
8	NC							

Position number	J1
Connector type	SMD PIN HEADER 2X5PIN-1.27MM-180 (KM51215624)
Name	JTAG Interface
From/To	From System to DC-DC



Pin	Signal name	Type	Voltage Min	Voltage Max	Current Min	Current Max	Low max High min	Notes
1	3.3V	Power		3.3VDC		<0.5mA		
2	JTMS	Signal		3.3VDC		<0.5mA		
3	GND	Power		3.3VDC		<0.5mA		
4	JTCK	Signal		3.3VDC		<0.5mA		
5	GND	Power		3.3VDC		<0.5mA		
6	NC							
7	NC							
8	NC							
9	GND	Power		3.3VDC		<0.5mA		
10	RST	Signal		3.3VDC		<0.5mA		

Position number	XD1
Connector type	DB9 FEMALE CONNECTOR (KM1334288)
Name	DEBUG CONEECTOR
From/To	RCU TO USB DEBUG CONNECTOR



Note: Pin number is not same

Pin	Signal name	Type	Voltage Min	Voltage Max	Current Min	Current Max	Low max High min	Notes
1	GND	SUPPLY						
2	RXD	INPUT						
3	TXD	OUTPUT						
4	NC							

5	GND	SUPPLY						
6	NC							
7	NC							
8	NC							
9	NC							

Position number	XO1	 Note: Pin number is not same
Connector type	FEMALE PIN HEADER 1X20PIN-2.54MM-180 (KM51621405)	
Name	LCD DISPLAY CONNECTOR	
From/To	DC-DC TO 2*16 DISPLAY MODULE	

Pin	Signal name	Type	Voltage Min	Voltage Max	Current Min	Current Max	Low max High min	Notes
1	VSS	SUPPLY	0VDC		1mA	1.5mA		
2	+3.3V_DISPLAY	SUPPLY		3.3V	1mA	1.5mA		
3	VO	SUPPLY			1mA	1.5mA		
4	3.3V_RS	OUTPUT		3.3V	1mA	1.5mA		
5	3.3V_RW	OUTPUT		3.3V	1mA	1.5mA		
6	3.3V_EN	OUTPUT		3.3V	1mA	1.5mA		
7	NC							
8	NC							
9	NC							
10	NC							
11	3.3V_DB4	OUTPUT		3.3V	1mA	1.5mA		
12	3.3V_DB5	OUTPUT		3.3V	1mA	1.5mA		
13	3.3V_DB6	OUTPUT		3.3V	1mA	1.5mA		
14	3.3V_DB7	OUTPUT		3.3V	1mA	1.5mA		
15	+3V3_DISPLAY	SUPPLY		3.3V		40mA		
16	B/L -	SUPPLY		3.3V		40mA		

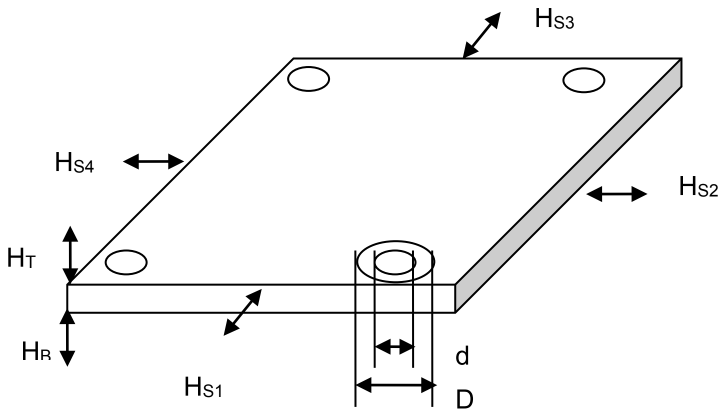
3.2 MECHANICAL INTERFACE

3.2.1 PCB LAYOUT DRAWING

- Boar size 83.2 x 46 mm

3.2.2 ALLOWED FIXING METHOD

<table>
<tr><th>Allowed fixing method:</th><th>Symbol</th><th colspan="2">Min</th><th colspan="2">Typical</th><th colspan="2">Max</th><th>Unit</th></tr>
<tr><td>Hole diameter (inner)</td><td>d</td><td colspan="2"></td><td colspan="2">3.2</td><td colspan="2"></td><td>mm</td></tr>
<tr><td>Diameter of the fixing part (outer)</td><td>D</td><td colspan="2"></td><td colspan="2"></td><td colspan="2">7.5</td><td>mm</td></tr>
<tr><td>Height from the bottom (to metal)</td><td>H_B</td><td colspan="2">6</td><td colspan="2"></td><td colspan="2"></td><td>mm</td></tr>
<tr><td>Height from the top (to metal)</td><td>H_T</td><td colspan="2">6</td><td colspan="2"></td><td colspan="2"></td><td>mm</td></tr>
<tr><td>Distance from the side (to metal)</td><td>H_{S1}</td><td colspan="2">2</td><td colspan="2"></td><td colspan="2"></td><td>mm</td></tr>
<tr><td>Distance from the side (to metal)</td><td>H_{S2}</td><td colspan="2">2</td><td colspan="2"></td><td colspan="2"></td><td>mm</td></tr>
<tr><td>Distance from the side (to metal)</td><td>H_{S3}</td><td colspan="2">2</td><td colspan="2"></td><td colspan="2"></td><td>mm</td></tr>
<tr><td>Distance from the side (to metal)</td><td>H_{S4}</td><td colspan="2">2</td><td colspan="2"></td><td colspan="2"></td><td>mm</td></tr>
<tr><td>Allowed material of the fixing part</td><td>-</td><td colspan="6">Plastic or metal</td><td>-</td></tr>
<tr><td>Allowed mounting positions</td><td></td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td><td>Yes</td><td>No</td><td>-</td></tr>
<tr><td>Other limits</td><td></td><td colspan="6"></td><td></td></tr>
</table>



3.3 USER INTERFACE

3.3.1 LED'S

LED position number	**Color**	**Name**	**Notes**
D6	Green	+3.3V OK	
D*	Red	Change Board/Error	

3.3.2 KEYPAD SWITCHES

DC-DC board is provided with 3 switches to navigate through the menu. SMD switches with 100mA and 42V rating is used. UP, DOWN, ENTER are the selections available to navigate with these switches.

3.3.3 FUSES

3.3.4 PARAMETERS

Not Applicable

3.4 DIAGNOSTICS

In DC-DC board below LED's are used for Diagnostics

LED Position Number	Glowing	Not Glowing
D6	3.3V is present	3.3V is not present
D*	Fault in board	No Fault in board

3.5 REQUIREMENTS FOR SOFTWARE

Not Applicable

3.6 DESIGN FOR TESTABILITY (DFT)

Not Applicable

3.7 INSTALLATION

Not Applicable

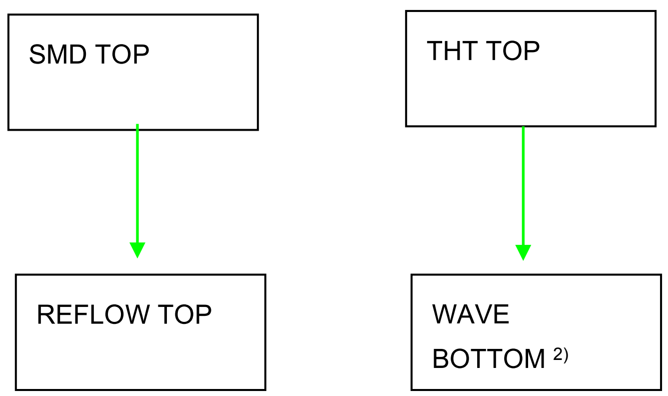
3.8 MAINTENANCE

3.9 MANUFACTURING PROCESS

3.9.1 PCB STRUCTURE AND COMPONENT TECHNOLOGY

Description	Method / Value	Note
Soldering process, top side	Reflow	
Soldering process, bottom side	Wave	
Component placement, top side	SMD + TH	
Component placement, bottom side	No	
Coating	Not needed	
PCB material	FR4	
Number of layers	2 or 4	Based on routing complexity
Resistor/Capacitor side	0402,0603,0805	
Logic family	HCMOS	
Voltage levels	380VDC,36V, 15V,12V 5V	

Traditional FR4 processes (green recommended, red not recommended):



1) When THT components will be soldered in reflow only, the components must be specified for reflow

2) Wave solder and reflow are alternative for each other

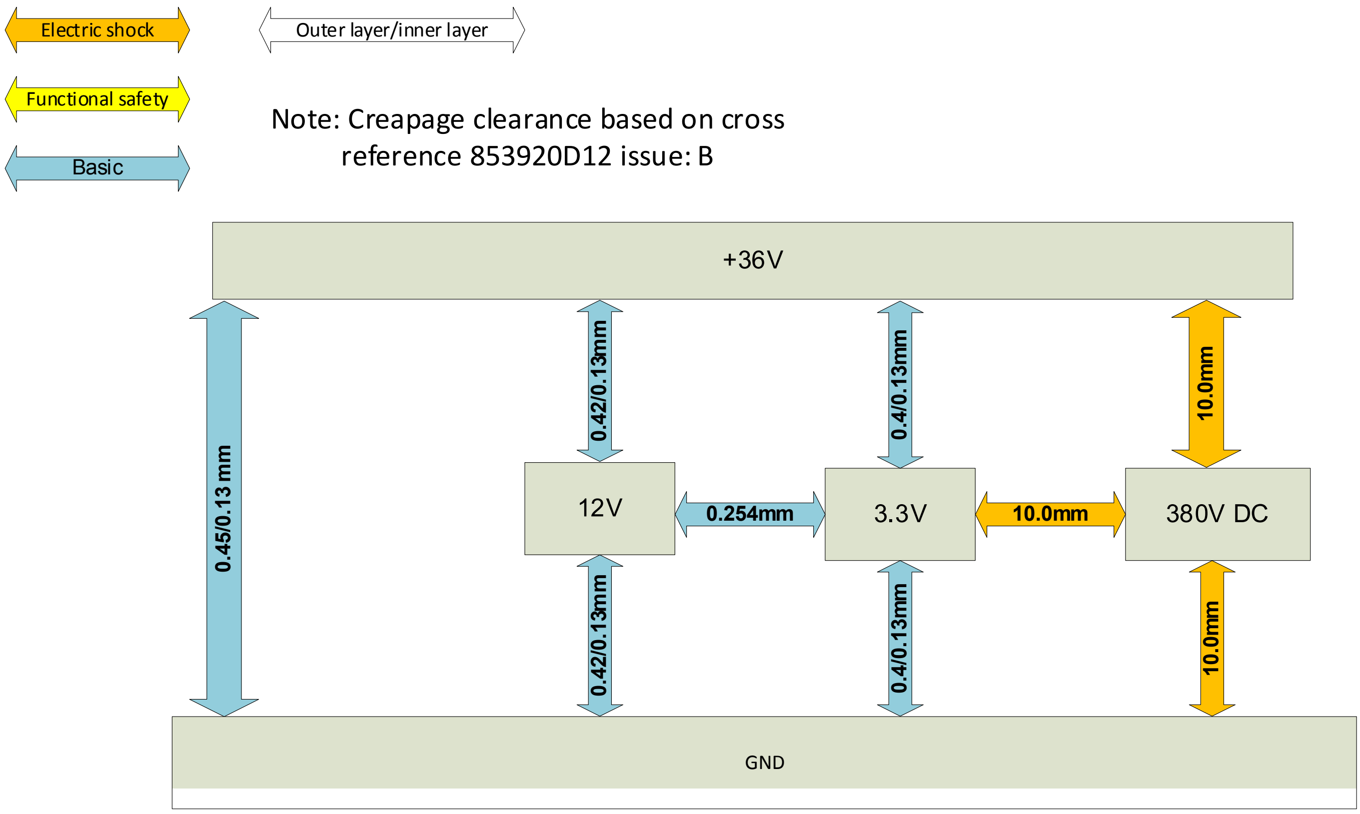
4 LAYOUT INSTRUCTIONS

Refer Section 5.2 STANDARDS AND REQUIREMENTS

4.1 SAFETY

- EN81-1, EN81-2: Safety rules for the construction and installation of lifts
- IEC 60950: Safety information technology equipment
- IEC 60664-1: Insulation coordination for equipment within low-voltage systems
- IEC 61558-1: Safety of power transformers, power supply units and similar
- CAN/CSA-B44.1-96 / ASME-A17.5-1996: Elevator and escalator electrical equipment, public safety
- Cross reference to clearances 853920D12

4.2 CLEARANCES AND CREEPAGE DISTANCES



4.3 EMC

- Emission EN 12015
- Immunity EN 12016

4.4 ENVIRONMENTAL REQUIREMENTS

See LAB-01.01.021. If different requirement, list details here.

4.5 RELIABILITY

- Reliability at design lifetime
  - Lifetime >15 years
  - Probability to not fail during lifetime 0,9 (no maintenance during this time)
- Early Failure rate during warranty time <0,002 (probability to fail)

4.6 PROTECTION CATEGORY

- IP21 (with mechanics)

5 COMPATIBILITY

- spare parts / replacements

6 SECOND SOURCES

- Needed changes/actions due to second source components

7 OTHER RELATED ANALYSIS AND TEST RESULTS

7.1 ANALYSIS RESULTS

- Process and design FMEA found in 51674123D13 (TBD)
- Stress analysis found in 51674123D14 (TBD)

7.2 TEST RESULTS

- Test results found in 51674123D11 (TBD)

8 TECHNICAL DATA

Parameter	Min	Typical	Max	Unit
Supply voltage	36	36	42	VDC
Supply current				mA
AC supply voltage				VAC
AC supply frequency				Hz
Operating temperature	-20		+66	ºC
Storing temperature	-40		+70	ºC
Humidity			36	g/m³
Operating altitude (above sea level)			4000	M
Mechanical dimensions Width Height Depth			 83.2 100 46	 mm mm mm
Weight		400		g

9 APPROVALS AND VERSION HISTORY

Compiled by: Praveen B
Checked by: Venkatesan S
Approved by: Rajkumar S

Issue	Date	Description of Change	Ref CR	Approved By
-	2021-02-11	First Draft	-	Rajkumar S
			-	

DRAFT